

Notes: Fama & French (JF, 2010)

Luck versus Skill in the Cross-Section of Mutual Fund Returns

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Mutual fund sample	2
2.2	Net and gross returns	2
2.3	Assets-under-management groups	3
2.4	Factor returns	3
3	Models and empirical specifications	3
3.1	Factor regression	3
3.2	Annualized alpha	4
3.3	Precision-adjusted alpha	4
3.4	Equilibrium accounting	4
3.5	Bootstrap null: no true alpha	4
3.6	Bootstrap CDF comparison	5
3.7	Likelihood measure	5
3.8	Injected-alpha simulations	5
4	Identification and empirical design	5
4.1	What identifies skill versus luck	5
4.2	Why joint sampling matters	5
4.3	Sample-selection logic	6
4.4	What the design cannot fully solve	6
5	Tables and figures	7
5.1	Table I + Table II: factors and aggregate active-fund performance	7
5.2	Table III + Figures 1–2: actual versus luck-only distributions	7
5.3	Table IV: how much true skill dispersion fits the gross-return tails?	8
5.4	Table V + Table AI: comparison with Kosowski et al. and CAPM robustness	9
6	Short summary	11
7	Conclusion	11
7.1	Core conclusion	11
7.2	Economic intuition	11
7.3	Takeaway	11

1 Big picture

- **Question.** Do active U.S. equity mutual fund managers have skill, or are the extreme winners and losers mostly the result of luck?
- **Core framework.** Performance is measured by the intercept α in factor regressions using the Fama–French three-factor model and the Carhart four-factor model.
- **Aggregate result.** The value-weight portfolio of actively managed equity funds is close to the market portfolio. Before expenses, aggregate active-fund α is close to zero; after expenses, net α is negative by about the cost of active management.
- **Equilibrium accounting.** If passive investors earn zero abnormal returns before costs, then active investing is a zero-sum game before costs and a negative-sum game after costs.
- **Main statistical problem.** With thousands of funds, some will look very good or very bad by chance. The task is to separate genuine skill from sampling luck.
- **Main empirical method.** The authors use bootstrap simulations of the cross-section of fund returns. Simulated fund returns preserve the empirical dependence structure of fund and factor returns, but true α is set to zero.
- **Net-return conclusion.** For investors, the evidence is bleak: very few funds produce benchmark-adjusted expected returns sufficient to cover their costs.
- **Gross-return conclusion.** If expense ratios are added back, the tails suggest both inferior and superior managers. But the estimated dispersion of true gross α is modest.
- **Estimated skill dispersion.** A normal cross-section of true gross α with mean zero and annual standard deviation around 1.25% fits the tails reasonably well.
- **Comparison to Kosowski et al.** Kosowski et al. find stronger evidence of skill. Fama and French argue that the difference is driven by their independent bootstrap design, fund inclusion rules, survival bias, and sample period.
- **Economic takeaway.** Active management may contain skill in the tails, but for most investors active mutual funds do not produce positive net abnormal returns. Costs largely absorb the benefits of skill.

2 Data and measurement

2.1 Mutual fund sample

- The sample comes from the CRSP mutual fund database.
- The authors include U.S. equity mutual funds and exclude index funds.
- Different share classes of the same fund are combined into a single fund using value weights.
- The main sample period is January 1984 through September 2006.
- The focus on the post-1983 period avoids important selection bias from older CRSP data, when many funds reported only annual returns.

Interpretation: The sample is designed to evaluate active management while reducing mechanical biases from share classes, index funds, and incomplete return reporting.

2.2 Net and gross returns

- Net returns are the returns received by investors.
- Gross returns are constructed as net returns plus 1/12 of the annual expense ratio.

- When a fund’s expense ratio is missing, the authors impute it using actively managed funds with similar assets under management.
- Gross returns do not fully restore all trading costs, but they reverse the cost included in the expense ratio.

Interpretation: The distinction is central. Net returns ask whether investors benefit. Gross returns ask whether managers have stock-picking ability before the expense-ratio cost paid by investors.

2.3 Assets-under-management groups

The paper sorts funds into three AUM samples:

- funds above \$5 million in AUM;
- funds above \$250 million in AUM;
- funds above \$1 billion in AUM.

In the main Table III sample there are:

3,156 funds in the \$5 million group,

1,422 funds in the \$250 million group,

660 funds in the \$1 billion group.

Interpretation: The smaller-AUM group gives a broader sample but noisier histories. The larger-AUM groups focus on economically more important funds and tend to have longer histories.

2.4 Factor returns

The paper uses:

- $R_{Mt} - R_{ft}$: market excess return;
- SMB_t : small-minus-big size factor;
- HML_t : high-minus-low book-to-market factor;
- MOM_t : momentum factor constructed like HML but using prior returns.

For January 1984 to September 2006:

- the market premium averages 0.64% per month;
- SMB averages 0.03% per month;
- HML averages 0.40% per month;
- MOM averages 0.79% per month.

Interpretation: Momentum has the largest average factor return in the period, but the paper finds that adding momentum does not dramatically change the main mutual-fund performance conclusions.

3 Models and empirical specifications

3.1 Factor regression

The core monthly regression is

$$R_{it} - R_{ft} = a_i + b_i(R_{Mt} - R_{ft}) + s_iSMB_t + h_iHML_t + m_iMOM_t + e_{it}. \quad (1)$$

- The full regression is the Carhart four-factor model.
- Dropping MOM_t gives the Fama–French three-factor model.
- Dropping SMB_t , HML_t , and MOM_t gives the CAPM robustness test.
- The intercept a_i is the estimate of fund i 's monthly abnormal return.

Interpretation: A positive true intercept means the fund produces returns above comparable passive benchmark exposures. A negative intercept means the fund underperforms its passive clone.

3.2 Annualized alpha

The paper reports annualized abnormal returns as

$$12a_i. \quad (2)$$

Interpretation: Since returns are monthly, multiplying the intercept by 12 converts monthly abnormal return into an annual percentage. The approximation is sufficient for small monthly returns.

3.3 Precision-adjusted alpha

For each fund, the paper focuses on

$$t(\alpha_i) = \frac{\hat{a}_i}{SE(\hat{a}_i)}. \quad (3)$$

Interpretation: The $t(\alpha)$ statistic accounts for precision. A short-lived fund with a large estimated alpha but high standard error is not treated the same as a long-lived fund with the same point estimate and a smaller standard error.

3.4 Equilibrium accounting

Let the value-weight market portfolio be divided into passive and active portfolios. If passive investing earns zero gross abnormal return relative to the benchmarks, then before costs:

$$\alpha_{\text{active, aggregate}} = 0. \quad (4)$$

After costs:

$$\alpha_{\text{active, net}} = -\text{active management costs}. \quad (5)$$

Interpretation: This is the arithmetic of active management. In aggregate, active investors cannot all beat passive benchmarks before costs; after costs they must underperform.

3.5 Bootstrap null: no true alpha

For each fund, the bootstrap starts from adjusted returns that remove the estimated abnormal return:

$$R_{it}^0 = R_{it} - \hat{a}_i. \quad (6)$$

A simulation run samples months with replacement, using the same month draw for all funds and factor returns.

Interpretation: The simulated returns preserve the empirical cross-correlation of funds and factors, but impose the null that every fund has true $\alpha = 0$.

3.6 Bootstrap CDF comparison

For each simulation run and each AUM group, the authors estimate the same benchmark regression and form the cross-sectional CDF of $t(\alpha)$ estimates. They compare:

$$t_p^{Actual}(\alpha) \quad \text{versus} \quad E_{\text{sim}} [t_p^{Sim}(\alpha)], \quad (7)$$

where p is a percentile of the cross-sectional distribution.

Interpretation: If actual left-tail $t(\alpha)$ values are far below the simulated left tail, there is evidence of truly inferior managers. If actual right-tail values are far above the simulated right tail, there is evidence of truly superior managers.

3.7 Likelihood measure

The tables report

$$\% < Act = \Pr_{\text{sim}} (t_p^{Sim}(\alpha) < t_p^{Actual}(\alpha)). \quad (8)$$

Interpretation: A very low value in the left tail means actual bad performers are worse than luck alone predicts. A very high value in the right tail means actual good performers are better than luck alone predicts.

3.8 Injected-alpha simulations

To infer dispersion in true skill, the authors inject a true α into simulated gross returns:

$$\alpha_i^{true} \sim N(0, \sigma^2), \quad (9)$$

where σ is the annual standard deviation of true α .

They examine σ values such as

$$0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00 \quad \text{percent per year.}$$

Interpretation: The question is how much cross-sectional true skill dispersion is needed to make the simulated tails look like the actual gross-return tails.

4 Identification and empirical design

4.1 What identifies skill versus luck

- The paper does not rely on short-term winner persistence.
- It uses long fund return histories to estimate individual alphas.
- It then asks whether the cross-section of estimated $t(\alpha)$ values is more extreme than a luck-only bootstrap distribution.
- Skill is inferred from systematic deviations of actual percentile values from simulated percentile values.

Interpretation: The method directly confronts the multiple-testing problem: in a large cross-section, some funds will appear extreme by chance.

4.2 Why joint sampling matters

- A simulation run uses the same random sample of months for all funds.
- This preserves cross-sectional correlation in fund returns.
- It also preserves joint movement in factor returns and fund residuals.
- Independent fund-level bootstraps miss these dependencies and can overstate the evidence of skill.

Interpretation: The joint bootstrap is a central difference between Fama–French and Kosowski et al. It makes the simulated extremes more realistic.

4.3 Sample-selection logic

- Funds enter the main simulation after crossing an AUM threshold.
- A fund needs at least eight months of returns after crossing the threshold.
- This reduces survival bias relative to requiring long survival histories.
- Kosowski et al. require survival for more than 60 months, which Fama and French argue induces stronger survival bias.

Interpretation: The inclusion rule matters because high early returns by tiny incubated funds can enter the record only if the fund survives and becomes visible.

4.4 What the design cannot fully solve

- The bootstrap samples months independently, so it does not preserve return autocorrelation.
- Time-varying fund loadings are not modeled.
- Gross returns add back expense ratios but do not fully adjust for all trading costs.
- Factor models may miss relevant benchmark exposures.
- The paper tests long-run fund histories and does not give a simple implementable rule for selecting future winners.

Interpretation: The paper’s strength is not that it solves every benchmark problem. Its strength is that it gives a conservative cross-sectional test of whether extreme performance is more than luck.

5 Tables and figures

This section keeps the compact visual style used in the prior notes. Adjacent original tables and figures are grouped to save space and improve reading speed. All visuals are directly cropped from the paper.

5.1 Table I + Table II: factors and aggregate active-fund performance

Read: Table I.

- The factor means during 1984–2006 are: market 0.64% per month, *SMB* 0.03%, *HML* 0.40%, and *MOM* 0.79%.
- Momentum has the highest average return and strongest *t*-statistic among the factor returns.
- The size premium is essentially zero in this sample.

Read: Table II.

- EW net alpha is around -0.92 to -1.11% per year across CAPM, three-factor, and four-factor models.
- VW net alpha is around -0.81 to -1.13% per year.
- Gross alphas are close to zero, especially for the value-weight portfolio.
- Market loadings are close to one, and nonmarket factor loadings are small in the VW portfolio.

Economic message: In aggregate, active mutual funds look like the market before expenses and underperform by roughly their costs after expenses.

Table I
Summary Statistics for Monthly Explanatory Returns for the Three-Factor and Four-Factor Models

R_M is the return on a value-weight market portfolio of NYSE, Amex, and NASDAQ stocks, and *R_f* is the 1-month Treasury bill rate. The construction of *SMB* and *HML* follows Fama and French (1993). At the end of June of each year *k*, we sort stocks into two size groups. Small includes NY Amex, and NASDAQ stocks with June market capitalization below the NYSE median and Big includes stocks with market cap above the NYSE median. We also sort stocks into three book-to-market equity (*B/M*) groups, Growth (NYSE, Amex, and NASDAQ) stocks in the bottom 30% of *B/M*, Neutral (middle 40% of NYSE *B/M*), and Value (top 30% of NYSE *B/M*). Book equity is for the fiscal year ending in calendar year *k*–1, an market cap in *B/M* is for the end of December of *k*–1. The intersection of the (independent) size and *B/M* sorts produces six value-weight portfolios refreshed at the end of June each year. The size return, *SMB_t*, is the simple average of the month *t* returns on the three Small stock portfolios minus the average of the returns on the three Big stock portfolios. The value-growth return, *HML_t*, is the simple average of the returns on the two Value portfolios minus the average of the returns on the two Growth portfolios. The momentum return, *MOM_t*, is defined like *HML_t*, except that we use prior return rather than *B/M* and the momentum sort is refreshed monthly rather than annually. At the end of each month *t*–1 we sort NYSE stocks on the average of the 11 months of returns to the end of month *t*–2. (Dropping the return for month *t*–1 is common in the momentum literature use the 30th and 70th NYSE percentiles to assign NYSE, Amex, and NASDAQ stocks to Low, Medium, and High momentum groups. The intersection of the size sort for the most recent June and the independent momentum sort produces six value-weight portfolios, refreshed monthly. The momentum return, *MOM_t*, is the simple average of the month *t* returns on the two High momentum portfolios minus the average of the returns on the two Low momentum portfolios. The table shows the average monthly return, the standard deviation of monthly returns, and the *t*-statistic for the average monthly return. The period is January 1984 through September 2006.

	Average Return				Standard Deviation				<i>t</i> -statistic			
	<i>R_M</i> – <i>R_f</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>	<i>R_M</i> – <i>R_f</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>	<i>R_M</i> – <i>R_f</i>	<i>SMB</i>	<i>HML</i>	<i>MOM</i>
1984–2006	0.64	0.03	0.40	0.79	4.36	3.38	3.17	4.35	2.42	0.13	2.10	2.10

Table I: summary statistics for benchmark factors.

Table II
Intercepts and Slopes in Variants of Regression (1) for Equal-Weight (EW) and Value-Weight (VW) Portfolios of Actively Managed Mutual Funds

The table shows the annualized intercepts ($12 \times a$) and *t*-statistics for the intercepts (*t*(Coef)) for the CAPM, three-factor, and four-factor versions of regression (1) estimated on equal-weight (EW) and value-weight (VW) net and gross returns on the portfolios of actively managed mutual funds in our sample. The table also shows the regression slopes (*b*, *s*, *h*, and *m*, for *R_M*–*R_f*, *SMB*, *HML*, and *MOM*, respectively), *t*-statistics for the slopes, and the regression *R*², all of which are the same to two decimals for gross and net returns. For the market slope, *t*(Coef) tests whether *b* is different from 1.0. Net returns are those received by investors. Gross returns are net returns plus 1/12th of a fund's expense ratio for the year. When a fund's expense ratio for a year is missing, we assume it is the same as other actively managed funds with similar assets under management (AUM). The period is January 1984 through September 2006. On average there are 1,308 funds and their average AUM is \$648.0 million.

	12 × <i>a</i>		<i>b</i>	<i>s</i>	<i>h</i>	<i>m</i>	<i>R</i> ²
	Net	Gross					
EW Returns							
<i>Coef</i>	−1.11	0.18	1.01				0.96
<i>t</i> (<i>Coef</i>)	−1.80	0.31	1.12				
<i>Coef</i>	−0.93	0.36	0.98	0.18	−0.00		0.98
<i>t</i> (<i>Coef</i>)	−2.13	0.85	−1.78	16.09	−0.24		
<i>Coef</i>	−0.92	0.39	0.98	0.18	−0.00	−0.00	0.98
<i>t</i> (<i>Coef</i>)	−2.05	0.90	−1.78	16.01	−0.25	−0.14	
VW Returns							
<i>Coef</i>	−1.13	−0.18	0.99				0.99
<i>t</i> (<i>Coef</i>)	−3.03	−0.49	−2.10				
<i>Coef</i>	−0.81	0.13	0.96	0.07	−0.03		0.99
<i>t</i> (<i>Coef</i>)	−2.50	0.40	−5.42	7.96	−3.22		
<i>Coef</i>	−1.00	−0.05	0.97	0.07	−0.03	0.02	0.99
<i>t</i> (<i>Coef</i>)	−3.02	−0.15	−5.03	7.78	−3.03	2.60	

Table II: aggregate active-fund regressions.

5.2 Table III + Figures 1–2: actual versus luck-only distributions

Read: Table III.

- For net returns, actual *t*(α) values are far below simulation averages in the left tail.
- For net returns, the right tail provides little evidence that many managers cover their costs.
- For gross returns, adding back expenses pushes the actual right tail to the right of the simulated right tail.
- The gross-return tails indicate both inferior and superior managers before expenses.

Read: Figures 1–2.

- Figure 1 shows the actual CDF for net returns lies left of the simulated CDF over much of the distribution.
- Figure 2 shows the gross-return CDF crosses the simulated CDF: the left tail is worse, while the right tail is better.
- The figures are for the \$5 million AUM group and the three-factor benchmark.

Economic message: Costs are the difference between the two messages. Investors mostly see negative net abnormal returns; before expenses there is evidence of both good and bad managers in the tails.

**Percentiles of $t(\alpha)$ Estimates for Actual and Simulated Fund Returns:
January 1984 to September 2006**

The table shows values of $t(\alpha)$ at selected percentiles (Pct) of the distribution of $t(\alpha)$ estimates for actual (Act) net and gross fund returns. The table also shows the percent of the 10,000 simulation runs that produce lower values of $t(\alpha)$ at the selected percentiles than those observed for actual fund returns (% < Act). Sim is the average value of $t(\alpha)$ at the selected percentiles from the simulations. The period is January 1984 to September 2006 and results are shown for the three- and four-factor models for the \$5 million, \$250 million, and \$1 billion AUM fund groups. There are 3,156 funds in the \$5 million group, 1,422 in the \$250 million group, and 660 in the \$1 billion group.

Pct	5 Million			250 Million			1 Billion		
	Sim	Act	% < Act	Sim	Act	% < Act	Sim	Act	% < Act
3-Factor Net Returns									
1	-2.50	-3.87	0.08	-2.45	-3.87	0.10	-2.39	-4.39	0.01
2	-2.17	-3.42	0.06	-2.13	-3.38	0.13	-2.09	-3.55	0.09
3	-1.97	-3.15	0.07	-1.94	-3.15	0.12	-1.91	-3.36	0.07
4	-1.83	-2.99	0.06	-1.80	-3.04	0.10	-1.78	-3.16	0.07
5	-1.71	-2.84	0.08	-1.69	-2.91	0.10	-1.67	-2.99	0.10
10	-1.32	-2.34	0.05	-1.31	-2.37	0.10	-1.30	-2.53	0.08
20	-0.87	-1.74	0.03	-0.86	-1.87	0.04	-0.86	-1.98	0.03
30	-0.54	-1.27	0.06	-0.54	-1.41	0.06	-0.54	-1.59	0.02
40	-0.26	-0.92	0.05	-0.27	-1.03	0.07	-0.27	-1.19	0.02
50	-0.01	-0.62	0.04	-0.01	-0.71	0.06	-0.01	-0.82	0.03
60	0.25	-0.29	0.11	0.25	-0.39	0.19	0.24	-0.51	0.05
70	0.52	0.08	0.51	0.52	-0.08	0.25	0.52	-0.20	0.08
80	0.85	0.50	3.20	0.84	0.37	1.68	0.84	0.25	0.85
90	1.30	1.01	8.17	1.29	0.89	5.19	1.28	0.82	4.81
95	1.68	1.54	30.55	1.66	1.36	14.17	1.64	1.34	17.73
96	1.80	1.71	40.06	1.76	1.49	17.24	1.74	1.52	26.33
97	1.94	1.91	49.35	1.90	1.69	25.92	1.87	1.79	42.86
98	2.13	2.17	58.70	2.08	1.90	30.43	2.04	2.02	50.07
99	2.45	2.47	57.42	2.36	2.29	43.92	2.31	2.40	63.11
4-Factor Net Returns									
1	-2.55	-3.94	0.04	-2.47	-3.94	0.08	-2.40	-4.22	0.01
2	-2.20	-3.43	0.04	-2.14	-3.43	0.09	-2.09	-3.48	0.08
3	-2.00	-3.08	0.13	-1.95	-3.07	0.25	-1.91	-3.11	0.23
4	-1.85	-2.88	0.13	-1.80	-2.88	0.22	-1.77	-2.95	0.21
5	-1.73	-2.74	0.12	-1.69	-2.78	0.18	-1.66	-2.86	0.14
10	-1.33	-2.23	0.14	-1.30	-2.34	0.14	-1.29	-2.48	0.07
20	-0.86	-1.67	0.10	-0.85	-1.80	0.11	-0.84	-1.96	0.05
30	-0.53	-1.25	0.12	-0.52	-1.39	0.10	-0.52	-1.54	0.04
40	-0.25	-0.88	0.21	-0.25	-1.04	0.14	-0.25	-1.23	0.05
50	0.01	-0.60	0.18	0.01	-0.76	0.11	0.01	-0.87	0.07
60	0.26	-0.29	0.25	0.27	-0.42	0.29	0.26	-0.49	0.19
70	0.54	0.02	0.37	0.54	-0.13	0.24	0.54	-0.18	0.24
80	0.87	0.44	1.76	0.86	0.27	0.72	0.86	0.17	0.45
90	1.33	1.04	10.62	1.31	0.86	4.40	1.30	0.86	7.07
95	1.72	1.53	23.82	1.69	1.37	14.35	1.67	1.31	14.13
96	1.84	1.67	28.21	1.80	1.51	18.23	1.78	1.45	17.16
97	1.99	1.84	31.30	1.94	1.65	18.62	1.91	1.57	17.05
98	2.19	2.09	39.12	2.12	1.79	15.57	2.08	1.76	18.86
99	2.52	2.40	36.96	2.42	2.22	29.88	2.36	2.26	42.00

(continued)

Table III—Continued

Pct	5 Million			250 Million			1 Billion		
	Sim	Act	% < Act	Sim	Act	% < Act	Sim	Act	% < Act
3-Factor Gross Returns									
1	-2.49	-3.07	4.11	-2.45	-3.16	3.16	-2.39	-3.29	1.88
2	-2.17	-2.68	4.79	-2.13	-2.67	6.01	-2.09	-2.70	5.64
3	-1.97	-2.48	4.20	-1.94	-2.51	4.47	-1.91	-2.51	5.12
4	-1.83	-2.31	4.41	-1.80	-2.35	4.68	-1.78	-2.33	5.77
5	-1.71	-2.19	4.15	-1.69	-2.18	5.99	-1.67	-2.18	6.52
10	-1.32	-1.72	5.75	-1.31	-1.77	5.94	-1.30	-1.86	4.15
20	-0.87	-1.10	13.61	-0.86	-1.24	7.18	-0.86	-1.43	2.52
30	-0.54	-0.71	20.03	-0.54	-0.79	15.10	-0.54	-1.00	4.28
40	-0.26	-0.36	29.74	-0.27	-0.43	23.84	-0.27	-0.59	10.25
50	-0.01	-0.06	38.87	-0.01	-0.15	26.28	-0.01	-0.28	13.48
60	0.25	0.28	56.05	0.25	0.14	31.47	0.24	0.05	21.21
70	0.52	0.63	71.81	0.52	0.48	43.62	0.52	0.35	26.70
80	0.85	1.06	85.21	0.84	0.88	58.14	0.84	0.79	44.31
90	1.30	1.59	90.01	1.29	1.41	69.39	1.28	1.34	60.63
95	1.68	2.04	92.10	1.66	1.81	72.89	1.64	1.78	70.37
96	1.80	2.20	93.73	1.76	1.93	73.44	1.74	1.96	77.00
97	1.94	2.44	95.97	1.90	2.19	84.36	1.87	2.22	85.47
98	2.13	2.72	97.29	2.08	2.47	89.30	2.04	2.37	83.72
99	2.45	3.03	96.66	2.36	2.83	90.95	2.31	2.97	94.63
4-Factor Gross Returns									
1	-2.55	-3.06	5.49	-2.47	-3.02	6.72	-2.40	-3.34	1.67
2	-2.20	-2.71	4.99	-2.14	-2.63	7.84	-2.09	-2.48	14.14
3	-2.00	-2.46	5.46	-1.95	-2.43	7.33	-1.91	-2.40	8.43
4	-1.85	-2.27	6.39	-1.80	-2.33	5.73	-1.77	-2.25	8.66
5	-1.73	-2.11	7.71	-1.69	-2.12	8.62	-1.66	-2.11	9.52
10	-1.33	-1.62	12.27	-1.30	-1.71	8.63	-1.29	-1.85	4.69
20	-0.86	-1.09	16.23	-0.85	-1.19	11.13	-0.84	-1.34	5.29
30	-0.53	-0.65	28.46	-0.52	-0.75	19.76	-0.52	-0.92	8.75
40	-0.25	-0.33	35.43	-0.25	-0.45	22.31	-0.25	-0.57	12.54
50	0.01	-0.02	44.53	0.01	-0.16	26.29	0.01	-0.29	14.40
60	0.26	0.28	53.17	0.27	0.09	25.86	0.26	0.05	22.48
70	0.54	0.62	64.90	0.54	0.48	43.11	0.54	0.36	27.78
80	0.87	0.98	70.19	0.86	0.85	50.07	0.86	0.82	47.07
90	1.33	1.58	84.76	1.31	1.36	58.66	1.30	1.41	65.72
95	1.72	2.05	88.77	1.69	1.87	73.81	1.67	1.83	70.55
96	1.84	2.21	91.03	1.80	2.01	76.27	1.78	1.95	70.91
97	1.99	2.39	92.01	1.94	2.21	81.22	1.91	2.04	66.61
98	2.19	2.58	91.20	2.12	2.43	83.35	2.08	2.30	74.26
99	2.52	3.01	93.44	2.42	2.72	81.41	2.36	2.57	71.98

Table III, first page: net-return $t(\alpha)$ percentiles.

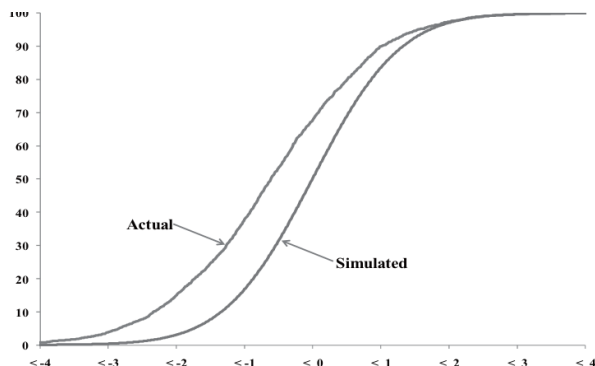


Figure 1. Simulated and actual cumulative density function of three-factor $t(\alpha)$ for net returns, 1984–2006.

Figure 1: net-return CDF, actual versus simulated.

Table III, continued: gross-return $t(\alpha)$ percentiles.

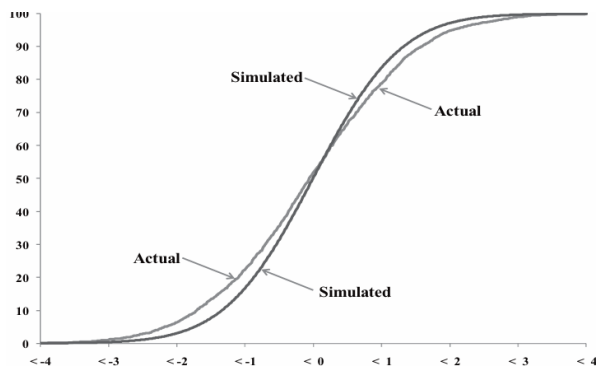


Figure 2. Simulated and actual cumulative density function of three-factor $t(\alpha)$ for gross returns, 1984–2006.

Figure 2: gross-return CDF, actual versus simulated.

5.3 Table IV: how much true skill dispersion fits the gross-return tails?

Read:

- Table IV injects normally distributed true gross α into the simulations.
- The annual standard deviation σ is varied from 0.50% to 2.00%.
- For the \$5 million AUM group, values around $\sigma = 1.25\%$ to 1.50% fit many tail percentiles.
- For the \$1 billion group, the implied right-tail dispersion is lower, often around 0.75% to 1.00%.
- The simulations have enough power to rule out values far from the fitted range.

Economic message: There is skill dispersion before costs, but it is modest. A standard deviation around 1.25% per year implies that very large true alphas are rare.

the average values of the $\hat{\alpha}(t)$ estimates at the same percentiles from the 10,000 simulations, for seven values of σ (the annual standard deviation of injected α). The seven columns of the table show, for each value of σ , the percent of the 10,000 simulation runs that produce lower $\hat{\alpha}(t)$ estimates at the selected percentiles than a fund returns. The period is January 1984 to September 2006 and results are shown for the three- and four-factor models for the \$5 million and \$1 billion AUM fund groups.

Pct	Actual $\hat{\alpha}(t)$	Average $\hat{\alpha}(t)$ from Simulations							Percent of Simulations below Actual						
		0.50	0.75	1.00	1.25	1.50	1.75	2.00	0.50	0.75	1.00	1.25	1.50	1.75	2.00
3-Factor α , AUM > \$5 Million															
1	-3.07	-2.63	-2.78	-2.99	-3.24	-3.54	-3.87	-4.23	7.46	15.74	36.37	69.29	92.13	99.00	9
2	-2.98	-2.27	-2.38	-2.52	-2.69	-2.89	-3.10	-3.34	4.03	13.67	26.30	49.41	76.25	93.77	9
3	-2.48	-2.06	-2.15	-2.27	-2.40	-2.55	-2.72	-2.91	6.55	10.94	19.32	35.16	57.73	80.85	9
4	-2.31	-1.91	-1.99	-2.08	-2.20	-2.33	-2.47	-2.62	6.85	10.63	17.71	30.94	49.77	71.63	8
5	-2.19	-1.78	-1.85	-1.94	-2.04	-2.16	-2.28	-2.41	6.36	9.68	15.54	26.25	41.56	61.44	8
10	-1.72	-1.37	-1.42	-1.48	-1.55	-1.62	-1.70	-1.78	7.86	10.82	15.39	22.37	32.27	44.75	5
90	1.59	1.35	1.40	1.46	1.53	1.60	1.68	1.76	86.35	81.64	74.23	64.06	51.23	36.75	2
95	2.04	1.75	1.83	1.92	2.02	2.13	2.26	2.39	88.27	82.46	72.10	58.14	38.57	18.02	
96	2.20	1.87	1.96	2.06	2.17	2.30	2.45	2.60	90.76	85.40	74.75	57.87	36.04	16.06	
97	2.44	2.03	2.12	2.23	2.37	2.53	2.70	2.88	93.73	89.76	80.72	63.35	38.59	15.13	
98	2.72	2.23	2.35	2.48	2.66	2.85	3.07	3.31	95.29	91.75	82.56	61.96	32.18	8.75	
99	3.03	2.58	2.74	2.95	3.20	3.49	3.82	4.18	93.48	85.84	63.90	29.57	5.82	0.41	
4-Factor α , AUM > \$5 Million															
1	-3.06	-2.69	-2.85	-3.06	-3.33	-3.63	-3.97	-4.34	10.99	22.26	47.07	78.71	95.82	99.64	9
2	-2.71	-2.31	-2.42	-2.57	-2.74	-2.94	-3.16	-3.41	8.36	14.61	28.65	52.04	78.54	94.44	9
3	-2.46	-2.09	-2.18	-2.30	-2.44	-2.60	-2.77	-2.96	8.82	14.08	25.32	42.58	66.01	86.35	9
4	-2.27	-1.93	-2.01	-2.11	-2.23	-2.36	-2.51	-2.66	9.85	15.06	24.90	39.73	60.10	80.07	9
5	-2.11	-1.80	-1.87	-1.96	-2.07	-2.18	-2.31	-2.45	11.46	16.87	28.22	39.83	57.77	76.75	9
10	-1.62	-1.38	-1.43	-1.49	-1.56	-1.64	-1.72	-1.80	16.05	21.02	37.97	57.70	80.46	93.12	7
90	1.58	1.38	1.43	1.50	1.56	1.64	1.72	1.80	79.81	74.21	66.46	55.96	43.49	30.25	1
95	2.05	1.80	1.87	1.96	2.07	2.18	2.31	2.44	83.71	76.60	66.13	50.01	31.99	15.33	
96	2.21	1.92	2.00	2.11	2.22	2.36	2.50	2.66	86.46	79.25	68.28	50.91	30.64	12.88	
97	2.39	2.08	2.17	2.29	2.43	2.59	2.76	2.95	87.10	79.52	66.70	46.44	24.15	7.38	
98	2.58	2.29	2.41	2.55	2.72	2.92	3.14	3.38	85.21	75.29	57.57	32.34	10.71	1.74	
99	3.01	2.66	2.81	3.02	3.28	3.58	3.91	4.27	88.10	75.85	49.98	19.30	3.25	0.19	
3-Factor α , AUM > \$1 Billion															
1	-3.29	-2.42	-2.54	-2.73	-2.99	-3.31	-3.68	-4.09	2.20	3.63	8.89	22.71	48.69	76.60	9
2	-2.70	-2.12	-2.21	-2.34	-2.52	-2.73	-2.98	-3.25	6.27	9.11	15.83	28.78	51.16	75.36	9
3	-2.51	-1.94	-2.01	-2.17	-2.34	-2.63	-2.84	-3.10	8.10	13.21	22.97	39.66	61.21	8	
4	-2.33	-1.80	-1.87	-1.97	-2.09	-2.23	-2.40	-2.57	6.43	8.75	13.73	22.41	36.16	55.38	7
5	-2.18	-1.69	-1.75	-1.84	-1.95	-2.08	-2.22	-2.37	7.42	9.81	14.45	22.65	35.12	51.86	7
10	-1.86	-1.32	-1.36	-1.42	-1.49	-1.58	-1.67	-1.77	4.46	5.54	7.88	11.48	17.22	25.15	3
90	1.34	1.29	1.34	1.40	1.47	1.55	1.64	1.74	58.48	52.67	43.86	34.00	23.16	13.78	
95	1.78	1.66	1.72	1.81	1.92	2.04	2.18	2.33	67.79	60.84	49.40	35.18	20.70	9.71	
96	1.96	1.77	1.83	1.93	2.05	2.19	2.35	2.53	74.69	68.02	56.45	40.71	24.10	11.07	
97	2.22	1.90	1.97	2.08	2.23	2.39	2.58	2.78	84.12	79.12	68.67	52.24	32.34	14.98	
98	2.37	2.07	2.16	2.29	2.46	2.67	2.90	3.16	82.02	75.12	61.59	41.62	21.60	6.82	
99	2.57	2.27	2.36	2.50	2.68	2.91	3.18	3.48	80.02	72.12	57.69	37.62	19.60	3.17	

Table IV, first page: injected- α simulations.

Pct	Actual ($\hat{\alpha}$)	Average ($\hat{\alpha}$) from Simulations							Percent of Simulations below Actual						
		0.50	0.75	1.00	1.25	1.50	1.75	2.00	0.50	0.75	1.00	1.25	1.50	1.75	2.00
4-Factor, AUM = \$1 Million															
1	-3.06	-2.69	-2.85	-3.06	-3.33	-3.63	-3.97	-4.34	10.99	22.26	47.07	78.71	95.82	99.64	9
2	-2.71	-2.31	-2.42	-2.57	-2.74	-2.94	-3.16	-3.41	8.36	14.61	28.65	52.40	78.54	94.44	9
3	-2.46	-2.06	-2.18	-2.30	-2.45	-2.62	-2.81	-3.02	8.82	14.08	28.40	42.68	66.35	86.35	9
4	-2.27	-1.93	-2.01	-2.11	-2.23	-2.36	-2.51	-2.66	9.85	16.66	24.90	39.73	60.00	80.00	9
5	-2.13	-1.79	-1.88	-1.97	-2.08	-2.20	-2.33	-2.47	10.76	17.45	25.69	39.73	57.17	75.75	9
10	-1.62	-1.38	-1.43	-1.49	-1.56	-1.64	-1.72	-1.80	10.65	16.62	25.97	37.70	49.49	63.12	9
90	1.58	1.38	1.43	1.50	1.56	1.64	1.72	1.80	78.81	74.21	66.46	55.96	43.66	30.25	1
95	1.95	1.80	1.87	1.96	2.09	2.24	2.40	2.59	88.71	81.33	68.48	54.99	41.85	26.13	1
96	2.21	1.92	2.00	2.11	2.22	2.36	2.50	2.66	64.66	49.75	35.68	26.08	16.91	30.64	12.88
97	2.39	2.08	2.17	2.29	2.43	2.59	2.76	2.95	87.10	79.52	66.70	46.44	24.15	7.38	
98	2.41	2.16	2.25	2.38	2.53	2.70	2.88	3.09	95.21	84.28	68.48	47.91	24.15	7.38	
99	2.61	2.31	2.66	2.82	3.02	3.28	3.58	3.91	92.88	88.10	75.55	49.98	19.30	3.25	0.19
4-Factor, AUM = \$1 Billion															
Pct	Actual ($\hat{\alpha}$)	0.50	0.75	1.00	1.25	1.50	1.75	2.00	0.50	0.75	1.00	1.25	1.50	1.75	2.00
4-Factor, AUM = \$1 Billion															
1	-3.34	-2.44	-2.56	-2.76	-3.03	-3.36	-3.74	-4.16	2.00	3.35	8.55	22.91	48.31	75.80	9
2	-2.48	-2.12	-2.22	-2.36	-2.54	-2.77	-3.02	-3.30	16.25	22.21	35.17	54.99	75.98	91.84	9
3	-2.40	-1.93	-2.01	-2.13	-2.28	-2.46	-2.66	-2.88	9.63	13.33	20.63	34.25	54.13	74.16	8
4	-2.25	-1.80	-1.87	-1.97	-2.10	-2.25	-2.42	-2.60	9.86	13.41	19.68	30.72	47.91	66.77	8
5	-2.11	-1.68	-1.75	-1.84	-1.96	-2.09	-2.24	-2.40	10.76	13.64	19.81	29.76	44.52	62.13	7
10	-1.85	-1.30	-1.35	-1.41	-1.49	-1.58	-1.67	-1.78	5.10	6.48	8.67	12.86	19.05	27.52	3
90	1.41	1.32	1.36	1.43	1.50	1.59	1.69	1.79	63.74	58.42	50.10	40.62	29.70	19.17	1
95	1.83	1.69	1.76	1.85	1.96	2.09	2.24	2.40	68.10	61.80	50.88	37.12	22.46	10.67	
96	1.95	1.80	1.87	1.97	2.10	2.25	2.41	2.59	68.50	61.33	49.38	34.72	19.04	7.98	
97	2.04	1.94	2.02	2.13	2.28	2.45	2.64	2.86	64.06	55.68	41.85	26.04	11.89	3.83	
98	2.30	2.12	2.21	2.35	2.53	2.74	2.98	3.25	71.76	62.55	47.39	28.62	11.89	3.45	
99	2.57	2.40	2.52	2.71	2.96	3.26	3.61	4.00	66.67	57.31	38.54	18.10	5.17	0.98	

Table IV, continued: injected- α simulations.

5.4 Table V + Table AI: comparison with Kosowski et al. and CAPM robustness

Read: Table V.

- Kosowski et al. find stronger evidence of positive performance in 1975–2002.
- Fama and French show that the evidence weakens when their joint bootstrap approach is used.
- Survival rules matter: requiring long fund survival can overstate the right tail of performance.
- With their exclusion rules, the 95th percentile $\% < Act$ is 68.32% rather than above 99%.

Read: Table AI.

- CAPM tests produce stronger evidence of apparent skill.
- The authors interpret this as a benchmark failure.
- CAPM misses size, value, and momentum tilts, especially value exposure.
- Once three- and four-factor models are used, the apparent CAPM skill largely weakens.

Economic message: Inferences about skill are highly sensitive to the bootstrap design and the benchmark model. Joint sampling and richer factor models make the evidence of skill much more limited.

Percentiles of Four-Factor $t(\alpha)$ for Actual and Simulated Fund Returns: 1975 to 2002

The table shows values of four-factor $t(\alpha)$ at selected percentiles (Pct) of the distribution of $t(\alpha)$ for actual (Act) net and gross fund returns for funds selected using the exclusion rules of Kosowski et al. (2006) and for funds in our \$5 million AUM group selected using our exclusion rules. The period is 1975 to 2002 (as in Kosowski et al. (2006)). The table also shows the fraction (%<Act) of the 10,000 simulation runs that produce lower values of $t(\alpha)$ at the selected percentiles than those observed for actual fund returns. Sim is the average value of $t(\alpha)$ at the selected percentiles from the simulations.

Pct	Kosowski et al. Exclusion Rules			Our Exclusion Rules		
	Sim	Act	%<Act	Sim	Act	%<Act
1	-2.48	-3.69	0.18	-2.46	-3.70	0.16
2	-2.16	-3.25	0.19	-2.14	-3.17	0.30
3	-1.96	-2.87	0.53	-1.95	-2.80	0.70
4	-1.82	-2.55	1.34	-1.80	-2.63	0.69
5	-1.70	-2.36	1.90	-1.69	-2.41	1.36
10	-1.31	-1.92	2.17	-1.30	-1.95	1.66
20	-0.85	-1.41	2.15	-0.85	-1.41	2.17
30	-0.52	-1.01	3.18	-0.52	-1.00	3.54
40	-0.25	-0.65	5.75	-0.24	-0.66	5.35
50	0.01	-0.33	9.19	0.01	-0.34	8.50
60	0.27	-0.02	12.20	0.27	-0.03	11.92
70	0.55	0.29	16.51	0.55	0.27	14.86
80	0.87	0.73	32.80	0.87	0.69	28.11
90	1.32	1.44	68.19	1.32	1.34	56.29
95	1.69	1.97	82.42	1.69	1.81	68.32
96	1.80	2.18	88.38	1.80	2.00	75.70
97	1.94	2.38	90.73	1.94	2.25	83.74
98	2.12	2.59	91.38	2.12	2.51	87.57
99	2.40	3.07	95.79	2.42	2.83	88.37

Table V: comparison with Kosowski et al.

Table AI
Percentiles of CAPM $t(\alpha)$ Estimates for Actual and Simulated Fund Returns

The table shows values of $t(\alpha)$ at selected percentiles (Pct) of the distribution of CAPM $t(\alpha)$ estimates for actual (Act) net and gross fund returns. The table also shows the percent of the 10,000 simulation runs that produce lower values of $t(\alpha)$ at the selected percentiles than those observed for actual fund returns (%<Act). Sim is the average value of $t(\alpha)$ at the selected percentiles from the simulations. The period is January 1984 to September 2006 and results are shown for the \$5 million, \$250 million, and \$1 billion AUM fund groups.

Pct	5 Million			250 Million			1 Billion		
	Sim	Act	%<Act	Sim	Act	%<Act	Sim	Act	%<Act
Net Returns									
1	-2.36	-3.72	0.25	-2.30	-3.70	0.40	-2.27	-4.10	0.08
2	-2.06	-3.28	0.45	-2.02	-3.29	0.58	-2.00	-3.50	0.24
3	-1.88	-3.00	0.64	-1.85	-3.02	0.79	-1.84	-3.29	0.23
4	-1.75	-2.84	0.62	-1.72	-2.92	0.65	-1.71	-3.18	0.19
5	-1.65	-2.69	0.74	-1.62	-2.76	0.77	-1.62	-3.00	0.27
10	-1.29	-2.16	1.08	-1.28	-2.18	1.64	-1.28	-2.47	0.46
20	-0.86	-1.48	1.93	-0.86	-1.58	1.98	-0.87	-1.79	0.70
30	-0.54	-1.05	2.09	-0.55	-1.11	2.30	-0.56	-1.35	0.44
40	-0.26	-0.65	3.84	-0.27	-0.75	2.50	-0.28	-0.88	0.48
50	0.00	-0.29	8.05	0.00	-0.36	5.23	-0.01	-0.46	1.29
60	0.26	0.08	20.79	0.26	0.06	19.86	0.26	-0.10	4.02
70	0.53	0.49	46.40	0.53	0.47	43.16	0.54	0.31	18.52
80	0.84	0.95	71.01	0.84	0.89	61.89	0.84	0.72	36.21
90	1.26	1.66	91.09	1.25	1.49	79.61	1.24	1.42	73.88
95	1.61	2.31	97.29	1.58	2.09	92.39	1.56	1.91	84.74
96	1.71	2.45	97.55	1.67	2.23	93.43	1.66	2.03	85.94
97	1.84	2.68	98.46	1.79	2.43	95.05	1.77	2.22	89.01
98	2.01	2.89	98.69	1.95	2.60	95.07	1.92	2.47	92.06
99	2.29	3.21	98.88	2.21	2.96	96.51	2.16	2.76	92.96
Gross Returns									
1	-2.36	-3.04	4.09	-2.30	-3.01	5.35	-2.27	-3.29	2.00
2	-2.06	-2.66	5.29	-2.02	-2.67	6.32	-2.00	-2.93	2.57
3	-1.88	-2.45	5.88	-1.85	-2.45	7.17	-1.84	-2.76	2.37
4	-1.75	-2.26	7.41	-1.72	-2.31	7.54	-1.71	-2.49	3.99
5	-1.65	-2.13	7.82	-1.62	-2.16	8.80	-1.62	-2.34	4.91
10	-1.29	-1.65	11.87	-1.28	-1.66	13.56	-1.28	-1.95	4.93
20	-0.86	-0.95	33.12	-0.86	-1.04	25.14	-0.87	-1.35	7.59
30	-0.54	-0.55	44.63	-0.55	-0.63	35.49	-0.56	-0.88	12.00
40	-0.26	-0.19	62.18	-0.27	-0.26	50.41	-0.28	-0.43	24.27
50	0.00	0.16	77.76	0.00	0.10	67.74	-0.01	-0.05	41.74
60	0.26	0.53	89.27	0.26	0.46	81.45	0.26	0.36	67.57
70	0.53	0.98	96.44	0.53	0.91	91.87	0.54	0.77	82.64
80	0.84	1.44	97.60	0.84	1.37	95.08	0.84	1.18	87.03
90	1.26	2.12	98.96	1.25	1.98	97.29	1.24	1.82	94.23
95	1.61	2.76	99.65	1.58	2.47	98.14	1.56	2.33	96.87
96	1.71	2.89	99.69	1.67	2.72	98.98	1.66	2.46	97.14
97	1.84	3.12	99.77	1.79	2.85	99.01	1.77	2.59	97.16
98	2.01	3.35	99.84	1.95	3.05	99.18	1.92	2.84	98.03
99	2.29	3.72	99.89	2.21	3.37	99.35	2.16	3.34	99.14

Table AI: CAPM bootstrap robustness.

6 Short summary

- Fama and French study whether mutual fund performance reflects luck or skill.
- The sample is actively managed U.S. equity mutual funds from CRSP, mainly 1984–2006.
- Fund performance is measured using CAPM, Fama–French three-factor, and Carhart four-factor alpha.
- The aggregate active-fund portfolio has gross alpha close to zero.
- Net alpha is negative by roughly the amount of expenses.
- This fits equilibrium accounting: active investing is zero-sum before costs and negative-sum after costs.
- To distinguish luck from skill, the authors use bootstrap simulations where true alpha is set to zero.
- The bootstrap samples the same months across funds to preserve cross-correlation in fund returns.
- Net-return tests show little evidence that many funds have enough skill to cover costs.
- Gross-return tests show evidence of both inferior and superior managers in the tails.
- A normal cross-section of true gross alpha with annual standard deviation around 1.25% fits the tails reasonably well.
- The evidence is weaker for large-AUM funds and stronger in CAPM tests only because CAPM misses factor tilts.
- The paper argues that Kosowski et al.’s stronger evidence is driven by independent bootstraps, survival rules, and sample-period differences.

7 Conclusion

7.1 Core conclusion

Fama and French conclude that active U.S. equity mutual funds, in aggregate, do not create abnormal value for investors. Before expenses, active funds as a group are close to zero alpha. After expenses, investors receive negative abnormal returns. The cross-section contains evidence of both superior and inferior ability before costs, but the amount of skill is modest and costs absorb much of its value.

7.2 Economic intuition

The intuition is the arithmetic of active management. If passive investors earn benchmark returns, active investors must collectively hold the rest of the market. Before costs, their aggregate abnormal return must net to zero. After costs, active investing becomes a negative-sum game. Therefore, any skilled managers must be offset by unskilled managers, and investors must also overcome fees and expenses.

7.3 Takeaway

The paper’s practical lesson is that extreme mutual fund performance is not reliable evidence of skill. In a large population, luck creates apparent winners. A joint bootstrap that preserves dependence across fund returns shows that few funds generate net abnormal returns sufficient to cover costs. Gross-return tails suggest some skill exists, but it is rare and economically modest.